

Magic Drops: Food 3D Printing of Colored Liquid Balls by Ultrasound Levitation

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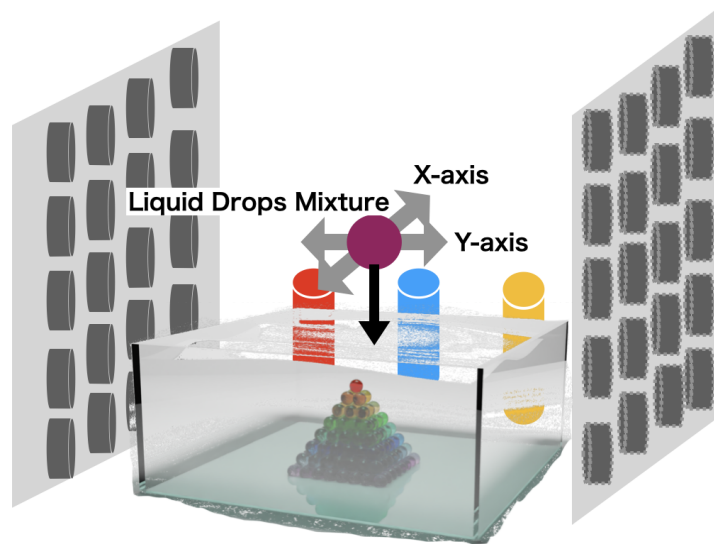


Figure 1: Overview of Magic Drops

ABSTRACT

We introduce the concept of "Magic Drops", the process which is all using ultrasound levitation to mix multiple liquid drops into a single one in the air and move it specified position and let it free fall below. For molecular gastronomy application, mixture drops with sodium alginate solution are free-fall into a container filled with calcium lactate solution. With this, drops encased in a calcium alginate film are formed in the container, these are edible and the color and flavor of mixture are also controlled through the process. We will also demonstrate stack these drops to create larger edible structure. Our novel mixture drop control technology has other potential applications such as painting techniques and drug development. Thus, we believe that this concept will become a new technology for mixing liquids in the future.

*Both authors contributed equally to this research.

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CCS CONCEPTS

• Human-centered computing → Human computer interaction(HCI).

KEYWORDS

Food 3D printer, Molecular gastronomy, Spherification, Ultrasound levitation

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1 INTRODUCTION

Food 3D printers are attracting attention as a device that can realize unprecedented food experiences and solve problems related to food. For example, there is research on chocolate that has different textures depending on its structure [3]. Another example is research to print something that looks like the real thing but is softer, so that people who cannot eat hard foods can enjoy their meals[1]. Most food 3D printers are based on the method of layering paste-like ingredients. However, we considered that there are two main issues with that method. The first is the difficulty in adjusting the moisture content. If the moisture content is too high, the material

may not stack and spread horizontally, while if the moisture content is too low, the nozzle may clog. Second, the range of color and taste expression is limited. In the case of the stacked type, filaments and nozzles must be prepared in advance for the required number of colors. Therefore, it is very time-consuming to try to achieve rich color and taste expression with the stacked type.

2 RELATED WORK

In this chapter, we present “nūfood”[2], which is related to the food 3D printing of liquid objects. This is a device that outputs flavors wrapped in a gel. Our work differs from the “nūfood” in two ways. The first is the way the drops are dropped. “nūfood” is motor-driven to move the XY coordinates. In contrast, our work uses ultrasound levitation to move the drops. The second difference is that our device can only output flavors that have been added beforehand, whereas our device can also mix colors.

3 SYSTEM AND IMPLEMENTATION

We propose Magic Drops an innovative ultrasound printing technique for printing liquid objects.

3.1 Concept

Magic Drops is the process of mixing multiple drops of liquid in mid-air by ultrasound levitation and falling them free to a designated location. Plastic balls, small animals have previously been made to float via ultrasound levitation[4, 5]. However, we thought that utilizing ultrasound levitation, liquids may be mixed without touching them.

3.2 Spherification

In molecular gastronomy, spherification is the process of forming drops out of liquid by enclosing it in a calcium alginate membrane. Calcium lactate ionizes in water to form calcium and lactate ions. Alginic acid and calcium ions combine to generate calcium alginate when sodium alginate is dissolved in water containing calcium ions. To create a calcium alginate membrane that is insoluble in water, this process is repeated. After some time, calcium alginate forms a gelatinous film around the sodium alginate solution, and drops start to form. Figure 2 shows the drops created by the authors.

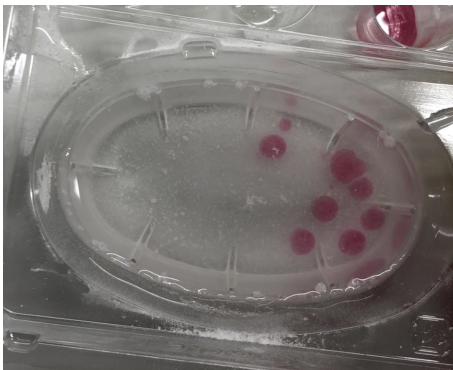


Figure 2: the drops tested by the authors

3.3 Food 3D Printing System

A sodium nitrate solution colored with food coloring is used in this device to simulate the falling liquid mentioned in the concept. The location where the combined liquid free falls is then prepared with a tray containing calcium lactate solution.

The device receives information on shape, color, and taste from the user. Based on the information provided and the spherification mentioned in Section 3.2, the device stacks the drops according to that information. For example, a colorful pyramid can be created as shown in Figure 1.

This device has three main features.

- It can create practically any color expression by outputting and combining the necessary colors from the magenta, cyan, and yellow tubes, respectively.
- Utilizing ultrasound levitation, move the liquid in a three-dimensional path as determined by coordinate data.
- It is possible to express characters by arranging drops, and three-dimensional expressions can be made by stacking drops.

By developing and testing this demonstration tool, we can verify the viability of the “Magic Drops” concept.

4 FUTURE WORK

This implementation focused on a food 3D printer that prints colored water enclosed in a membrane. However, Magic Drops concept can be extended to other uses. We propose two specific examples. The first is the picture dripping technique. Dripping is a technique of dripping paint onto a sheet of paper. This technique was used by Jackson Pollock, a leading American painter of small and medium-sized paintings. Dripping is a technique for enjoying coincidences and color changes. By applying the Magic Drops process, we can enjoy how coincidences are created when you drip paint on a designated spot on the canvas. The second is its application to chemical synthesis. When synthesizing chemicals, it is necessary to measure the weight of each sample and mix them in a test tube. However, with this device, a predetermined amount of chemicals can be output and dripped into a test tube placed below. Thus, we believe that this technology will be a flexible, gravity-independent means of printing in the future.

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